Statistical Decision Making
45-752

Models with Indicator Variables and Continuous Variables:
Modeling and Predicting the Prices of Automobile Tires

This file is named: 05_slides.pptx
Commands are in: 05_notebook.ipynb

1

Background

We now study an excellent application of regression by a Tepper alum.

Tire purchases are a very significant expense item for automobile manufacturers.

Our Tepper alum, working for an auto manufacturer, developed a statistical model for his company to predict prices of tires.

His motivations for developing the model were the following:

- Predict tire prices to aid the purchasing teams in negotiating tire prices with suppliers.
- For the automobile design team, predict of the cost of equipping a car with the type of tire that the design team is considering for a tire they are designing.

The company gave permission for our alum to share the data and statistical analysis with us.

A description of the variables in the data file is provided on the next slide.

2

2

Tire Application From Our Alum

Data Description: There are 61 observations, one for each of 61 different types of tires

Variable to Be Predicted: Price (Dollars per Tire)

Tire Characteristics:

Diameter: Inside diameter of the tire in inches

Ratio: Ratio of height of tire sidewall to width of tread, in percent.

The speed safety classification of each tire:

- R rated for normal highway driving
- H rated to 90 miles per hour
- V rated to 120 miles per hour
- Z rated to 150 miles per hour

Type of Exterior

- BW: Black Wall
- WW: White Wall
- OWL: Outside White Lettering

Terrain: equals 1 if tire is for all Terrain vehicle, 0 otherwise

Weight: weight of tire in pounds

Tread: tread width in millimeters

Volume: The number of tires of this type purchased by the company in the reference year (thousands)

3

3

Summary Statistics

The summary statistics for the variables are show here.

```
# Note: .transpose() transposes the summary
# statistics to a more compact form
Tire_Pricing.describe().transpose()
```

The statistics for Volume tell us that the company bought an average of 254,000 tires per purchase, with a maximum of 2.028 million and a minimum of 5,000.

We see that 70.5% of tires had speed rating for normal high way driving speeds.

	count	mean	std	min	25%	50%	75%	max
Obs	61.0	31.000000	17.752934	1.00	16.00	31.00	46.00	61.00
Price	61.0	99.687541	23.533104	59.82	78.27	100.35	117.66	151.56
Diameter	61.0	15.540984	0.807749	14.00	15.00	16.00	16.00	17.00
Ratio	61.0	64.836066	7.301097	45.00	60.00	65.00	70.00	75.00
Tread	61.0	222.868852	20.503232	185.00	215.00	225.00	235.00	265.00
Weight	61.0	25.180328	5.609570	16.20	21.50	24.30	27.70	40.80
Terrain	61.0	0.131148	0.340363	0.00	0.00	0.00	0.00	1.00
Volume	61.0	254.229508	387.708584	5.00	49.00	120.00	288.00	2028.00
R	61.0	0.704918	0.459865	0.00	0.00	1.00	1.00	1.00
H	61.0	0.114754	0.321370	0.00	0.00	0.00	0.00	1.00
V	61.0	0.098361	0.300273	0.00	0.00	0.00	0.00	1.00
Z	61.0	0.081967	0.276591	0.00	0.00	0.00	0.00	1.00
BW	61.0	0.950820	0.218039	0.00	1.00	1.00	1.00	1.00
WW	61.0	0.049180	0.218039	0.00	0.00	0.00	0.00	1.00
OWL	61.0	0.196721	0.400819	0.00	0.00	0.00	0.00	1.00

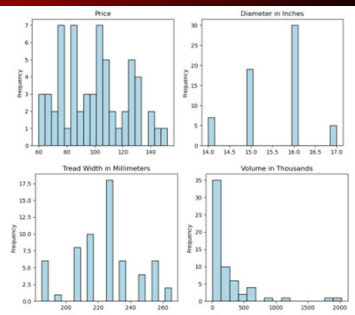
4

4

Histograms of Some Continuous Variables

The Volume variable is skewed to the right.

The remaining three variables are more symmetrically distributed.



5

Further Summary of The Data

Recall that the Volume variable is measured in thousands.

Issue the command:

```
# Total volume of tires purchases
total_volume = Tire_Pricing['Volume'].sum()
total_volume
```

- You will obtain 15,508. The company purchased 15.5 million tires during the year studied by our alum.
- Our alum's analysis illustrates the value of regression in studying a big-data application.

Issue the command:

```
# Total expenditures on tire purchases
total_expenditures = Tire_Pricing[['Price', 'Volume']].product(axis=1).sum()
total_expenditures
```

- You will obtain 1,409,866. The company spent $1.41 billion on tire purchases in the year studied by our alum.

Tires are a major purchase item for automobile companies!

6

6

The Regression Estimated by Our Alum

The regression estimated by our alum:

```
TireReg= tprstats.model("cs", 'Price~Diameter+Ratio+Terrain+Weight+Tread+Volume+WW+OWL+H+V+Z', Tire_Pricing)
TireReg.summary()
```

This model has multiple categories of indicator variables:

- Terrain has two categories, yes if all-terrain, no if not all-terrain.
- Exterior color has two categories, BW, WW
- Outside white letter has two categories: OWL, no OWL
- Speed rating has four categories (R,H,V,Z)

The reference chosen for each category is a matter of convenience and can easily be inferred from the above regression command:

- The reference category for all-terrain is a tire that is not all-terrain.
- Black wall is the most common type of exterior. It is natural to choose BW as the reference category for exterior as our alum did.
- The most common speed rating is a tire rated for normal highway driving, R. Hence, it is natural to make type R tire the reference among the four speed ratings: (R,H,V,Z)

The model also has multiple continuous variables: Diameter, Ratio, Weight, Tread, Volume.

7

7

Regression Results

Dep. Variable:	Price	**R-squared:**	0.933
Model:	OLS	**Adj. R-squared:**	0.918
No. Observations:	61	**F-statistic:**	106.2
Covariance Type:	HC1	**Prob (F-statistic):**	2.99e-30

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-27.2056	52.689	-0.516	0.608	-133.089	78.678
Diameter	5.3728	2.780	1.933	0.059	-0.213	10.959
Ratio	-0.6184	0.271	-2.284	0.027	-1.163	-0.074
Terrain	8.9199	3.203	2.785	0.008	2.483	15.357
Weight	1.8909	0.627	3.018	0.004	0.632	3.150
Tread	0.1305	0.167	0.782	0.438	-0.205	0.466
Volume	-0.0033	0.002	-1.775	0.082	-0.007	0.000
WW	4.4036	2.738	1.608	0.114	-1.099	9.907
OWL	4.0631	3.127	1.299	0.200	-2.222	10.348
H	8.0769	2.412	3.348	0.002	3.229	12.924
V	23.8924	4.204	5.684	0.000	15.445	32.340
Z	26.3225	5.772	4.561	0.000	14.724	37.921

Recall three **Key Criteria** for evaluating regression results:

Significance: Are the variables statistically significant?

Sign: Do the algebraic signs make sense for variables that are significant?

Magnitude: Are the magnitudes of the significant coefficients plausible?

- Use TireReg.scaled_coefficients() command to find the elasticity of price with respect to diameter. (It is .838.) Interpret the result.
- Is the coefficient of volume large enough to be managerially important?
- Should we keep an insignificant variable such as the Tread variable?

8

8

Is Volume Discount Large Enough to Be Managerially Important?

How can we assess whether the volume discount is managerially important?

- The coefficient of volume is -.0033.
- Average volume per purchase is 250 thousand tires.
  - How do we know this?
- Hence, volume discount **per tire** on an average purchase is 250*.0033=.825. This is $0.825.
- Hence, on a purchase of 250,000 tires, the volume discount savings would be the number of tires purchased time savings per tire:

  250,000*.825=$206,000

This is obviously large enough to be managerially important!

9

9

Thinking About The Model

WW, OWL, Tread are not very significant. Three possible reasons:

1) Those variables don't affect tire prices.
2) Those variables do affect tire prices but we don't have enough observations or variation to get precise estimates. (Verify that 3 tires have WW, 12 have OWL, none have both.)
3) The variables are highly correlated with each other making it difficult to distinguish their separate effects.

Should you keep or drop WW, OWL, or Tread?

- If you were in the industry, you would be able to judge whether there is an extra cost to OWL or WW and whether $4 is a reasonable estimate of the cost. (It seems reasonable to me.)
- Tread and weight are relatively highly correlated (.91). As an industry expert, you could judge whether to keep both. If the coefficients are reasonable in magnitude, there is no harm in retaining them (i.e., dropping one or the other will not make your prediction more precise).
- Our alum judged that it was best to retain these variables, and I agree.

10

10

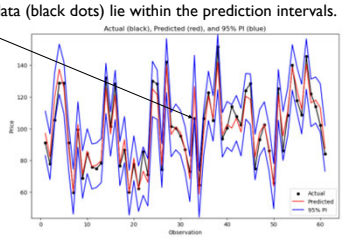
Plot Actual and Fitted Prices with Prediction Intervals

```
# Plot actual and fitted values
# The function uses 95% prediction intervals
TireReg.plot_actual_fitted()
```

We see that the fitted values (red) track the data (black dots) relatively well.

We also see that all but one of the data (black dots) lie within the prediction intervals.

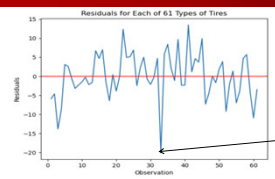
Looking at a plot of the residuals (next slide) will provide more information about outliers, if any.



11

11

Investigating The Residuals



See accompanying notebook for the commands to create this plot.

Let's investigate this large outlier. How can we do that?

We can find the observation number of this residual using the Python command:

```
# Find and view the index of the large negative residual
index = resids.argmin()
index
```

We find that this tire is observation 33 in our data.

Looking at the spreadsheet for the Tire_Pricing data, we see this is a Z tire.

12

12

Investigating The Outlier

Select and view data for all Z tires:

```
# Select all Z tires and view to see if there is anything unusual about this tire.
Ztires= Tire_Pricing.query('Z==1')
Ztires
```

	Obs	Price	Diameter	Ratio	Tread	Weight	Rating	Tirewall	Terrain	Volume	R	H	V	Z	BW	WW	OWL
4	5	128.82	16	60	215	24.0	Z	WW	0	27	0	0	0	1	0	1	0
23	24	130.11	16	55	225	23.6	Z	BW	0	16	0	0	0	1	1	0	0
26	27	142.02	17	45	245	24.1	Z	BW	0	120	0	0	0	1	1	0	0
32	33	106.53	16	50	215	23.5	Z	BW	0	20	0	0	0	1	1	0	0
37	38	151.56	17	45	245	25.1	Z	BW	0	50	0	0	0	1	1	0	0

Is there anything unusual about the outlier (Obs 33) compared to the other "Z" tires?

Compared to the other four Z tires, tire 33 does not seem unusual. It appears that the company simply negotiated a bargain on that purchase.

13

13

Predictions

Application: The design team in the company is anticipating using a tire with the following characteristics for the new car they are designing:

- The tire under consideration would have a diameter of 15, ratio of 65, weight of 25, and tread width of 225. The tire would be speed-rated for normal highway driving (R), would not be all terrain, would not have white walls or outside white lettering

If this tire configuration is adopted for the car, the company expects to purchase approximately 300,000 of the tires each year.

The designers would like to have an estimate of the price that they can expect to pay for the tire, and they would like a price range that they can be 75% sure of achieving when negotiating purchase.

14

14

Prediction and Sensitivity Analysis

To obtain the prediction:

```
Xnew = pandas.DataFrame({
    'Diameter': [15], 'Ratio': [65], 'Terrain': [0], 'Weight': [25], 'Tread': [225],   'WW': [0],
    'OWL': [0], 'H': [0], 'V': [0],  'Z': [0],  'Volume': [300]
})
TireReg.prediction_intervals(Xnew, alpha=0.25)
```

Predicted	Lower	Upper
88.845747	80.661586	97.029907

We obtain predicted price=$88.84 with 75% prediction interval ($80.66, $97.03).

Suppose we are unsure about retaining the Tread variable because it its relatively insignificant (p-value=.44). How can we investigate?

- Sensitivity analysis: Let's remove Tread from the regression model and re-estimate.
- Then redo the prediction.

Predicted	Lower	Upper
88.025207	79.862866	96.187548

We obtain predicted price =$88.02 with 75% prediction interval ($79.86,$96.19)

- We conclude our prediction is not very sensitive to whether we include the Tread variable.

Note that sensitivity analysis is an extremely valuable strategy for helping you prepare for possible criticisms when you prepare the presentation of your analysis. When in doubt about retaining one or more variables, try your analysis with and without those variables.

15

15

Functional Form

We can investigate functional form using a test called the Ramsey test.

We will study functional form in our next slide deck and provide an explanation of the Ramsey test.

We will then apply the Ramsey test to investigate the functional form chosen by our alum.

We will find that the functional form fits the data well.

16

16

Summary

We studied regression models with both continuous and indicator variables.

We reviewed interpretation of the coefficients of indicator variables.

We explained the meaning of reference category and saw that the choice of the reference category is largely a matter of convenience.

We studied an excellent application of multiple regression to aid managerial decision making.

As you go forward in future Tepper courses and your careers, follow the example of our alum, and keep an eye out for opportunities to apply stastical analysis to enhance managerial decision making.

17

17